

When an Initial Value Problem Meets a Boundary Value Problem: The Thomas-Fermi Story

Singular Origins, Infinite Domains, and Verification Certificates

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SIRPY · Problem 9 — BVP Solution with Asymptotic Behavior

An ongoing series. We are building a public library of problems SIRPY has solved — each entry is one problem, the exact input we handed the solver, the exact output it returned, and an honest read of what happened. No slides. No hand-waving. Numbers you can check.

Historical Background

The Thomas-Fermi model (1927) was one of the first attempts to describe multi-electron atoms using quantum mechanics. Llewellyn Thomas and Enrico Fermi independently developed this statistical model, treating electrons as a degenerate Fermi gas moving in a self-consistent electrostatic potential.

Despite its simplicity, the model captures several essential features of atomic structure:

- Electron density distributions
- Atomic screening effects
- Scaling laws for large atoms
- The universal Thomas-Fermi screening function

No known closed-form solution exists for the Thomas-Fermi equation, making numerical methods essential. The famous initial slope

$$y'(0) \approx -1.5880710226$$

is universal in the dimensionless neutral-atom formulation: every neutral atom corresponds to the same dimensionless solution, differing only through scaling transformations.

Why This Matters Today

Although modern density-functional theory (DFT) has replaced the Thomas-Fermi model for quantitative atomic calculations, the model remains important because:

1. It provides the leading asymptotic description of large atoms.
2. It serves as a benchmark for numerical methods on singular, semi-infinite domains.
3. Closely related equations appear in plasma physics, astrophysics, and condensed-matter theory.
4. Its mathematical structure (singular origin, nonlinear coupling, and boundary condition at infinity) appears in many modern scientific problems.

The Universal Nature of the Slope

The value

$$y'(0) \approx -1.5880710226$$

is remarkable because:

- It is universal for the neutral-atom Thomas-Fermi problem.
- It is determined entirely by the differential equation and the boundary condition at infinity.
- It emerges as an eigenvalue-like parameter of the boundary value problem.
- Computing it accurately requires solving a challenging nonlinear problem on a semi-infinite domain.

Physical Interpretation

The function $y(x)$ describes the dimensionless screening potential of the atom.

Near the nucleus ($x \rightarrow 0$):

- $y(0) = 1$
- $y'(0) \approx -1.5880710226$

Far from the nucleus ($x \rightarrow \infty$):

- $y(x) \rightarrow 0$
- $y(x) \sim 144/x^3$

The decay condition at infinity is precisely what selects the unique physical solution and determines the famous slope.

Historical Numerical Challenges

For decades the Thomas-Fermi equation served as a difficult benchmark for numerical analysis because:

- The coefficient $1/\sqrt{x}$ is singular at the origin.
- The physical boundary condition is imposed at infinity.
- The correct initial slope is unknown *a priori*.
- Small errors in the slope grow dramatically at large x .

As a result, the Thomas-Fermi slope became a classic test of numerical accuracy, with increasingly refined estimates appearing throughout the twentieth century.

Why I Like This Benchmark

Most ODE solvers are comfortable with smooth initial value problems.

The Thomas-Fermi equation is not one of them.

$$y'' = \frac{y^{3/2}}{\sqrt{x}}$$

The equation is singular at

$$x = 0.$$

The physically relevant solution satisfies

$$y(0) = 1, \quad y(\infty) = 0,$$

and the famous Thomas-Fermi slope is approximately

$$y'(0) \approx -1.588071.$$

This benchmark is interesting because it simultaneously contains:

- a singular origin,
- fractional-power behavior,
- an infinite-domain condition,

- a physically meaningful missing parameter,
- and a known critical slope.

The Initial Value Problem

We begin with the classical initial-value formulation

$$y'' = \frac{y^{3/2}}{\sqrt{x}}, \quad y(0) = 1, \quad y'(0) = -1.5880710226.$$

```
solve_ivp(
    order=2,
    f_str="y**(3/2)/sqrt(x)",
    x0=0,
    ic={0:1,1:-1.5880710226}
)
```

The Difficulty at $x = 0$

At first glance this problem appears singular:

$$\frac{1}{\sqrt{x}}$$

blows up at the origin.

Rather than pretending the singularity does not exist, SIRPY performs a Frobenius analysis.

The solver recognizes that the natural variable is

$$u = \sqrt{x}$$

which transforms the equation into a form with analytic coefficients.

The resulting indicial roots are

$\{0, 2\}$ ¹.

Most importantly:

the physical initial slope is not imposed artificially.

It appears naturally as one of the free coefficients of the Frobenius expansion.

This is exactly the behavior expected from the Thomas-Fermi problem.

Automatic March Without a User-Supplied Endpoint

The first experiment did not specify any terminal point.

```
solve_ivp(  
    order=2,  
    f_str="y**(3/2)/sqrt(x)",  
    x0=0,  
    ic={0:1, 1:-1.5880710226}  
)
```

Instead of requiring the user to specify a terminal point, SIRPY:

1. constructed a Frobenius start segment on $[0, 0.01]$ to handle the singular origin;
2. computed a reliable local radius estimate;
3. analytically continued the solution through successive Taylor segments;
4. selected a default horizon based on the measured series radius.

The solver reported:

¹**What are indicial roots?** In the Frobenius method, we assume a solution of the form $y(x) = x^r \sum_{n=0}^{\infty} a_n x^n$ near a regular singular point. The indicial equation determines the possible values of the exponent r . For the Thomas-Fermi equation, the roots $\{0, 2\}$ indicate that two solution branches exist near $x = 0$: the physical branch with $y(0) = 1$ (corresponding to $r = 0$) and a branch that vanishes at the origin (corresponding to $r = 2$). The boundary condition selects the physical branch, and the slope $y'(0)$ remains a free parameter determined by the far-field boundary condition.

Frobenius start:

[0, 0.01]

Analytic continuation:

[0.01, 2.95644]

21 series segments

The result:

Verified range:

[0, 2.95644]

21 analytic segments

Verification:

max residual:

1.74e-10

series-piece relative defect:

6.20e-15

overlap certificate:

1.69e-16

Interface continuity:

2.1e-11

Initial data:

machine precision

Visual verification residual 1.74e-10 [VERIFIED]

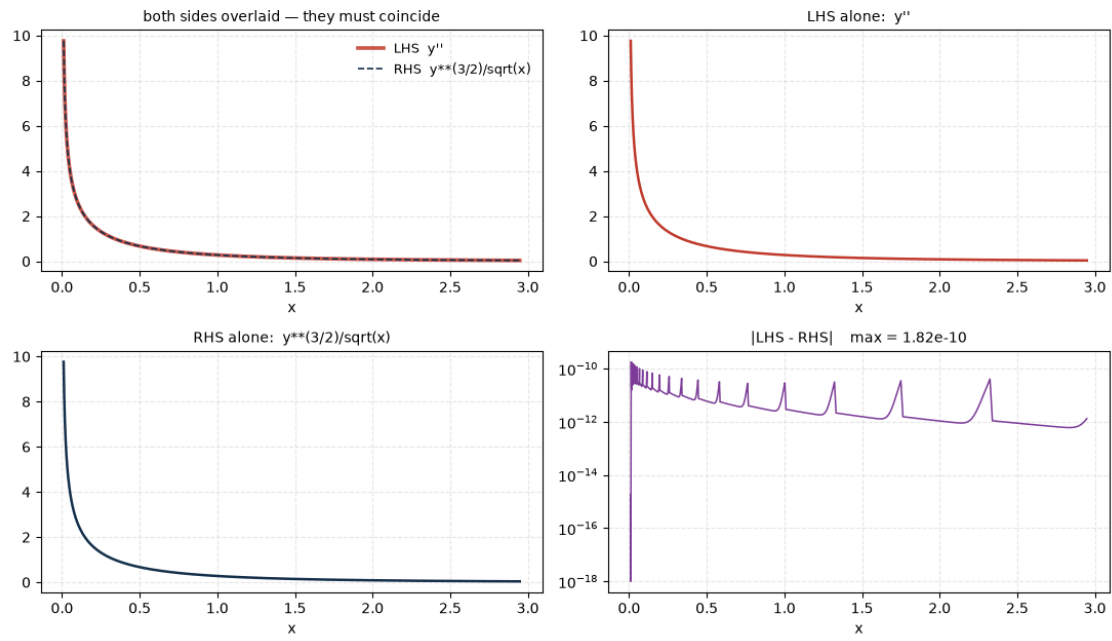


Figure 1: Verification BVP: both sides of the equation compared across the interval.

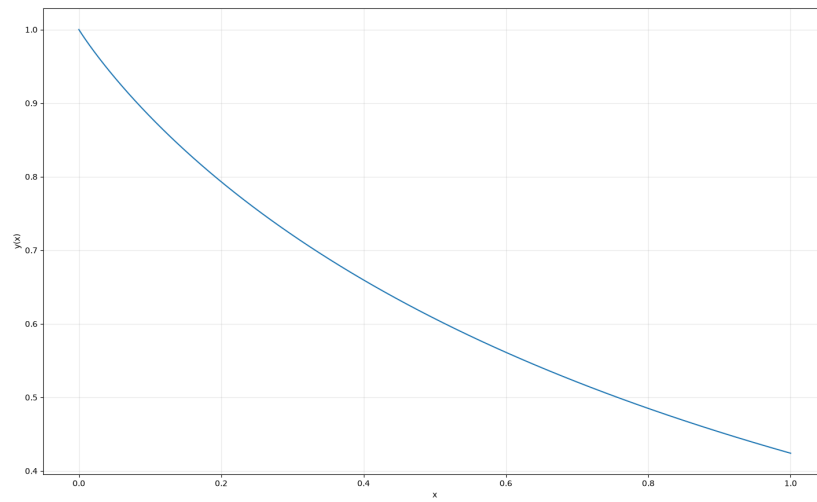


Figure 2: Solution Thimas-Fermi (BVP).

Extending the Same Solution to $x = 10$

The same IVP can be extended simply by providing a larger endpoint.

```
solve_ivp(  
    order=2,  
    f_str="y**(3/2)/sqrt(x)",  
    x0=0,  
    x1=10,  
    ic={0:1, 1:-1.5880710226}  
)
```

No reformulation is required.

This time SIRPY analytically continued all the way to

$$x = 10.$$

Results:

```
25 analytic segments
```

```
Equation residual:
```

```
1.74e-10
```

```
Interface continuity:
```

```
7.2e-11
```

An additional feature appeared:

```
nearest singularity estimate:
```

$$x \approx -0.00115063$$

This tells us where the nearest complex obstruction is expected to lie and therefore gives information about local series reliability.

Visual verification residual 1.74e-10 [VERIFIED]

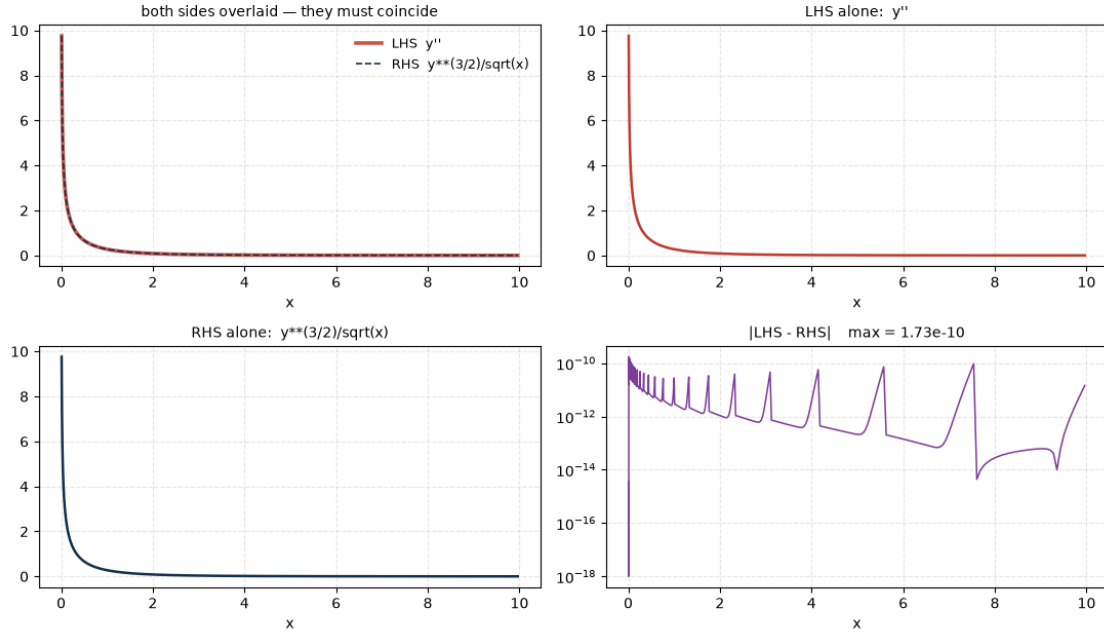


Figure 3: Verification BVP: both sides of the equation compared across the interval.

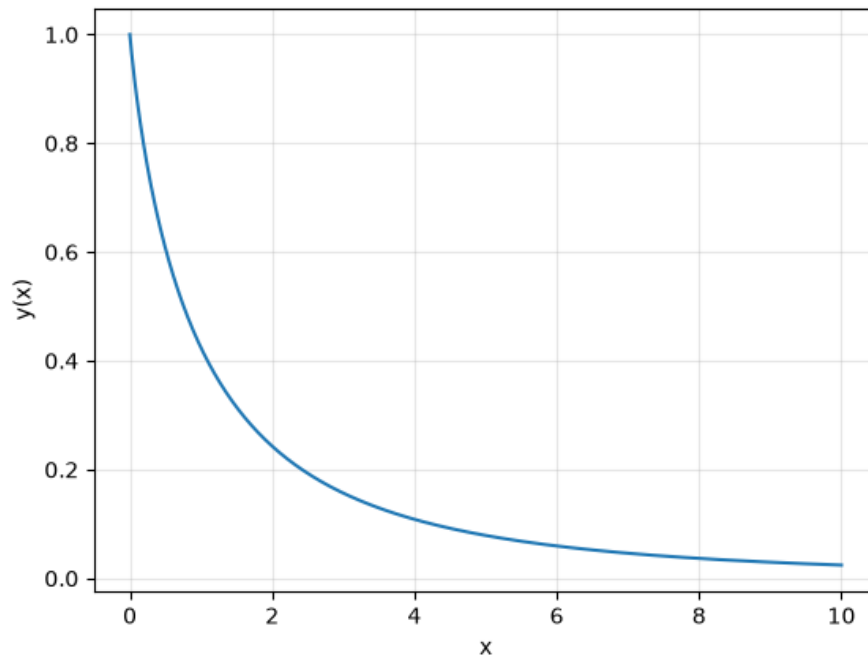


Figure 4: Solution Thimas-Fermi (BVP).

The Physical Boundary Value Problem

The more famous formulation is not an IVP.

The missing initial slope is unknown.

Instead we impose

$$y(0) = 1, \quad y(\infty) = 0.$$

```
solve_bvp(  
    order=2,  
    f_str="y**(3/2)/sqrt(x)",  
    x0=0,  
    x1=np.inf,  
    left_bc={0:1,1:"gamma"},  
    right_bc={0:0}  
)
```

Now the problem becomes:

Find the unique initial slope that causes the solution to decay to zero at infinity.

Handling Infinity

True infinity is impossible numerically.

Rather than silently truncating the problem, SIRPY reports its truncation analysis.

and then re-solved on successive truncation levels.

The resulting truncation certificate was:

$$\Delta = 1.10e - 04$$

meaning the inferred slope changed only slightly between truncation levels.

Recovering the Thomas-Fermi Slope

The solver found:

```
gamma = -1.58808570653
```

which closely matches the classical Thomas-Fermi value.

The endpoint shooting residual was:

```
1.13e-05
```

after enforcing the asymptotic condition.

Verification

After solving the BVP, SIRPY again verified the resulting solution.

```
Equation residual:
```

```
1.74e-10
```

```
Interface continuity:
```

```
1.1e-10
```

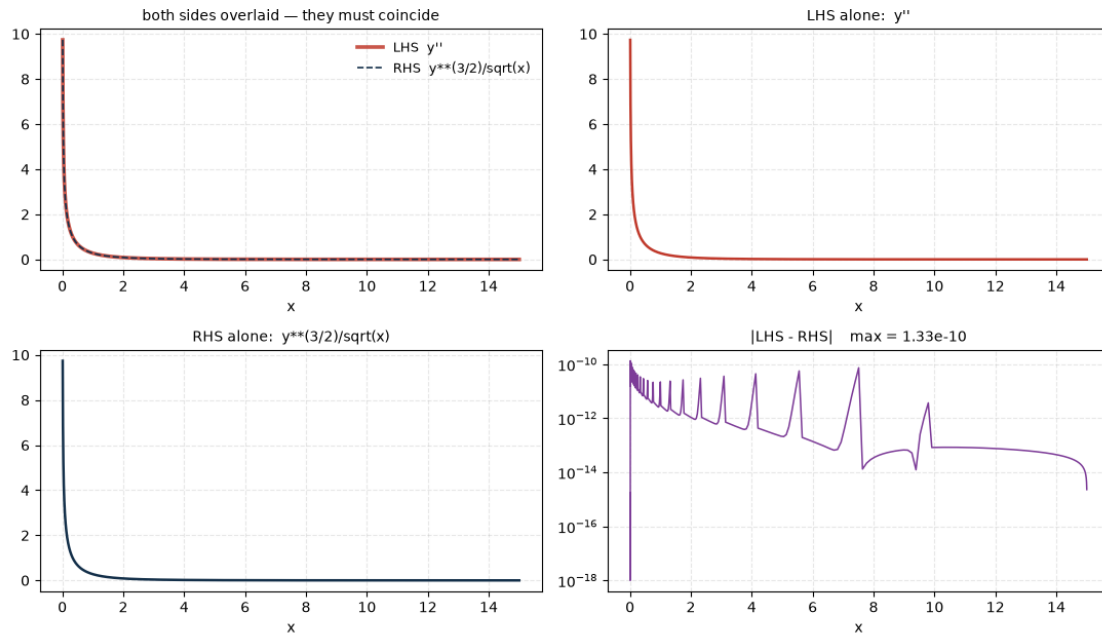
```
Initial condition:
```

```
machine precision
```

```
45 analytic segments
```

The entire solution remained represented as a chain of verified analytic Taylor pieces.

Visual verification residual $1.74\text{e-}10$ [VERIFIED]



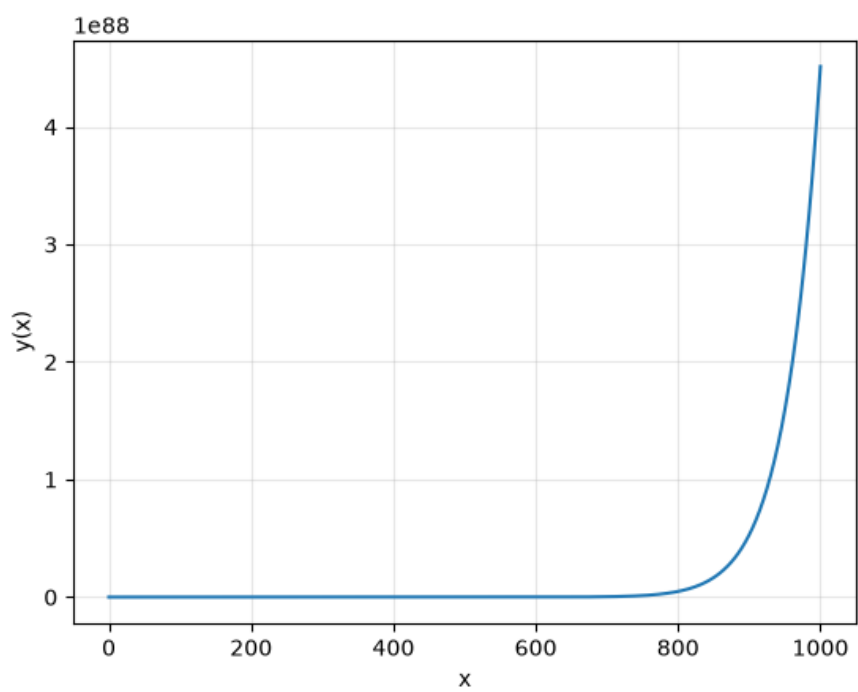


Figure 5: Solution Thimas-Fermi (BVP).

Why This Benchmark Matters

The Thomas-Fermi problem exercises nearly every difficult component of an ODE solver:

- singular origin,
- fractional powers,
- Frobenius expansions,
- analytic continuation,
- automatic horizon selection,
- infinite-domain boundary conditions,
- parameter identification,
- verification by substitution.

Many solvers can integrate the equation.

Far fewer can explain:

- why the singularity is harmless,
- how the regular branch is selected,
- how infinity is handled,
- how the slope is recovered,
- and why the final answer should be trusted.

Summary

Problem	Status
IVP (automatic horizon)	VERIFIED
IVP ($x = 10$)	VERIFIED
BVP (infinite boundary)	VERIFIED

Highlights:

Physical slope recovered:
-1.58808570653

Equation residual:
1.74e-10

```
Interface continuity:
```

```
1e-10
```

```
Infinite-domain truncation certificate:
```

```
provided
```

The Thomas-Fermi benchmark demonstrates something that I believe is increasingly important in scientific computing:

Solving the differential equation is only half the job.

The other half is explaining why the computed solution should be trusted.

The problem library

This entry is part of the SIRPY Problem Library — an ongoing, public catalog of mathematical systems solved and verified by SIRPY.

- **GitHub** — github.com/Mitra-Institute-of-Education-MIE/sirpy-problem-library
- **Web** — mitrainstituteofeducation.org/sirpy

Contact us

Contact: info@mitrainstituteofeducation.org · mitrainstituteofeducation.org/contact

Please only send problems you are free to share publicly.

SIRPY is in final validation prior to its first public release. Every line quoted above is verbatim from a single run of the solver; nothing is reproduced from memory.

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